

Software Requirements Specification (SRS)

Bean & Brew Smart Cafe Management System



Project Name: Smart Cafe Management, Online Ordering & Table
Reservation System

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1.Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the **Smart Cafe Management, Online Ordering & Table Reservation System**.

The system is designed to improve cafe operations by providing digital solutions for customer ordering, table reservations, order tracking, inventory management, supplier management, billing, and business reporting.

This document serves as a reference for understanding the system requirements and guiding the design, development, testing, and evaluation of the system.

1.2 Project Overview

The Smart Cafe Management System is a web-based cafe management solution developed for **Bean & Brew Cafe**.

The system provides different functionalities for customers and cafe staff/administrators. Customers can browse the menu, place orders, make table reservations, and track their orders. Cafe staff can manage orders, reservations, inventory, suppliers, staff information, and business reports.

The system aims to reduce manual work, minimize order errors, prevent double bookings, improve inventory visibility, and support better decision-making through reporting and analytics.

1.3 Document Scope

This document defines the requirements of the Smart Cafe Management System, including:

- Customer management
- Online menu browsing
- Online ordering
- Table reservations
- Order tracking
- Billing and invoice management
- Inventory management
- Supplier management

- Reporting and analytics
- User authentication and access control

2. Business Problem

Current Business Situation: The Smart Cafe is a growing cafe business that provides food and beverage services to customers. The business handles several day-to-day activities, including customer ordering, table reservations, billing and payments, inventory management, supplier coordination, and business reporting.

At present, several of these activities rely on manual processes and separate methods of recording and managing information. Customer orders and payments may require manual entry, while table reservations need to be monitored to ensure that tables are available at the required time. Inventory information also needs to be regularly updated and monitored to identify available and low-stock items.

As the number of customers and daily transactions increases, managing these activities manually becomes more difficult. Employees may need to spend additional time recording information, checking availability, updating stock, and preparing reports. This can increase the possibility of errors and delays in daily operations.

In addition, management requires accurate and timely information to understand sales performance, customer activity, product demand, inventory levels, and other operational activities. When information is maintained separately or reports are prepared manually, obtaining a complete view of the business can be time-consuming.

Therefore, the current business environment indicates a need to understand and improve the existing processes, information flow, and stakeholder needs before defining an appropriate digital solution.

Business Impact :

Problem	Business Impact
Manual ordering	Delays / order errors
Reservation issues	Customer dissatisfaction
Poor inventory visibility	Stock shortages
Manual billing	Transaction errors
Manual reporting	Slow decision-making

3. Project Objectives

The main objective of the Smart Cafe Management, Online Ordering & Table Reservation System is to improve the efficiency and effectiveness of cafe operations by reducing manual order processing and minimizing order-related errors. The system aims to enable customers to place orders and make table reservations digitally while reducing the possibility of duplicate reservations. It also aims to improve inventory visibility and stock monitoring, automate invoice and billing-related processes, and centralize cafe operational information within a single system. Furthermore, the system will provide role-based access to ensure that users can access functionalities according to their responsibilities. The system is also intended to generate useful business reports to support

management decision-making and improve overall customer convenience, service quality, and operational efficiency.

Measurable Objectives

Objective	Target
Reduce manual order entry	Reduce manual order handling by at least 50%
Reduce reservation conflicts	Prevent duplicate bookings through system validation
Improve inventory monitoring	Provide real-time stock information
Improve reporting	Generate sales and operational reports through the system
Improve customer convenience	Allow customers to order and reserve tables digitally

4. Stakeholder Analysis

Stakeholder	Role	Main Needs / Interests	Influence
Cafe Owner	Business decision maker	Revenue, performance, growth	High
Cafe Manager	Operations management	Orders, staff, inventory, reports	High
Customers	System users	Easy ordering, reservations, payments	High
Cashier	Order/payment handling	Fast and accurate transactions	Medium
Kitchen Staff	Order fulfilment	Clear and timely orders	Medium
Inventory Manager	Stock management	Accurate stock information	High
Suppliers	Supply products	Orders and communication	Medium
System Administrator	System management	Security, users, system maintenance	High

5. Scope

5.1 In Scope

The Bean & Brew Cafe Management System will include the following functions:

1. Customer Management

- Customer registration and login.
 - Customer profile management.
 - Viewing order history.
 - Managing customer feedback.
2. Menu Management
 - Adding, updating, and removing menu items.
 - Managing item prices and availability.
 - Allowing customers to view available menu items.
 3. Online Ordering
 - Customers can select food and beverages.
 - Customers can add items to a cart and place orders.
 - Staff can view and update order status.
 - Customers can track the status of their orders.
 4. Table Reservation Management
 - Customers can make table reservations.
 - Staff can view and manage reservations.
 - The system can maintain table availability.
 - Reservation status can be updated or cancelled.
 5. Billing and Payment Management
 - Automatic calculation of order totals.
 - Generation of invoices/receipts.
 - Recording payment information.
 - Tracking payment and billing status.
 6. Inventory Management
 - Recording available stock.
 - Updating stock quantities.
 - Monitoring stock levels.
 - Identifying low-stock items.
 - Recording stock-in and stock-out activities.
 7. Supplier Management
 - Maintaining supplier information.
 - Recording supplier-related stock purchases.
 - Tracking supplier and inventory information.
 8. Staff and User Management
 - Managing staff accounts.
 - Assigning appropriate system roles.
 - Controlling access based on user roles.
 9. Reporting and Analytics
 - Generating sales reports.
 - Generating order reports.
 - Generating inventory reports.
 - Generating reservation reports.
 - Providing management with operational information for decision-making.

10. Notifications

- Providing relevant order and reservation status notifications.
- Alerting staff about low-stock items where applicable.

5.2 Out of Scope

The following functions are not included in the current project scope:

- Payroll and employee salary management.
- Full accounting and financial management.
- Tax filing and government reporting.
- Delivery management using external delivery companies.
- Advanced AI-based demand forecasting.
- Integration with external banking systems.
- Integration with third-party food delivery platforms.
- Multi-branch management beyond the selected cafe operation.
- Advanced customer loyalty and rewards management.
- Automated purchasing directly from suppliers.

5.3 Scope Boundary

The project focuses on digitizing and integrating the cafe's core operational processes: customer management, menu management, ordering, reservations, billing, inventory, suppliers, staff management, and reporting.

The system is intended to reduce manual work, improve data accuracy, increase operational visibility, and support faster business decision-making.

6. Functional Requirements

Functional requirements define what the Bean & Brew Cafe Management System should do.

ID	Functional Requirement	Priority
FR-01	The system shall allow customers to register and log in using their account credentials.	High
FR-02	The system shall allow customers to view available menu items, prices, descriptions, and availability.	High
FR-03	The system shall allow customers to add, update, and remove items from their shopping cart.	High
FR-04	The system shall allow customers to place food and beverage orders digitally.	High
FR-05	The system shall allow staff to view, accept, update, and manage customer orders.	High
FR-06	The system shall allow customers to view the current status of their orders.	Medium
FR-07	The system shall allow customers to check table availability and make reservations.	High

FR-08	The system shall prevent the system from accepting conflicting reservations for the same table and time slot.	High
FR-09	The system shall allow staff to create, update, confirm, cancel, and manage reservations.	High
FR-10	The system shall automatically calculate the total amount of an order based on the selected items and quantities.	High
FR-11	The system shall generate an invoice or receipt for completed orders.	High
FR-12	The system shall record payment details and payment status for each transaction.	High
FR-13	The system shall allow authorized staff to add, update, and remove inventory items.	High
FR-14	The system shall automatically update stock quantities when relevant inventory transactions occur.	High
FR-15	The system shall identify and display items that reach their defined low-stock level.	High
FR-16	The system shall allow authorized users to maintain supplier information	Medium
FR-17	The system shall allow authorized users to manage staff accounts and system roles.	Medium
FR-18	The system shall generate sales, order, inventory, and reservation reports.	High
FR-19	The system shall allow customers to submit feedback about their cafe experience.	Medium
FR-20	The system shall provide relevant notifications for order, reservation, and low-stock status updates.	Medium

7. Non-Functional Requirements

Non-functional requirements define how well the system should perform and the quality standards it should maintain.

ID	Category	Non-Functional Requirement	Target
NFR-01	Performance	The system should respond to normal user requests within an acceptable time	≤ 3 seconds for 95% of normal requests
NFR-02	Availability	The system should be available during the cafe's normal operating hours.	$\geq 99\%$ availability during operating hours
NFR-03	Security	The system shall authenticate users before providing access to protected functions.	100% of protected functions require authentication
NFR-04	Authorization	The system shall restrict system functions according to user roles.	Role-based access control
NFR-05	Data Security	Passwords and sensitive user information shall be securely protected.	Passwords stored using secure hashing
NFR-06	Usability	The system should provide a simple and understandable interface for customers and staff.	Common tasks should require minimal steps
NFR-07	Reliability	The system should maintain accurate transaction and inventory data during normal operation.	No intentional loss of successfully saved transactions
NFR-08	Scalability	The system should be capable of supporting increasing numbers of customers, orders, and inventory records.	Support future data growth without major redesign
NFR-09	Maintainability	The system should use a structured and modular design to simplify maintenance and updates.	Modular components and documented code
NFR-10	Compatibility	The web-based system should operate correctly on commonly used modern browsers.	Chrome, Edge, and Firefox
NFR-11	Accuracy	The system should calculate order totals and maintain inventory quantities accurately.	100% accuracy for system calculations under defined rules

8. Business Rules

Business rules define the policies and conditions that control how the cafe operates and how the system should handle business activities.

ID	Business Rule
BR-01	A customer must be registered and logged in before placing an online order.
BR-02	Customers can only order menu items that are currently available
BR-03	The quantity of an ordered item must be greater than zero.
BR-04	An order must contain at least one menu item before it can be submitted.
BR-05	The system shall calculate the order total based on the item price and selected quantity.
BR-06	A table cannot be reserved by two customers for the same date and time period
BR-07	A reservation must contain a valid customer, table, date, and time.
BR-08	Customers may cancel a reservation only before the cafe's defined cancellation deadline.
BR-09	Only authorized staff can confirm, modify, or cancel customer reservations.
BR-10	Inventory quantities shall not be allowed to become negative.
BR-11	When stock reaches or falls below the defined minimum level, the item shall be identified as low stock.
BR-12	Only authorized staff can add, update, or remove inventory records.
BR-13	An invoice shall be generated for a completed order.
BR-14	A payment must be recorded against a valid order.
BR-15	Only authorized users can access management reports.
BR-16	Users shall only access functions permitted by their assigned system role.
BR-17	Cancelled orders shall not be treated as completed sales transactions.
BR-18	Completed transactions should not be deleted without appropriate authorization.

9. User Stories

User stories describe system requirements from the perspective of the users who interact with the system.

Customer User Stories

ID	User Story	Priority
US 01	As a Customer , I want to register an account, so that I can use the cafe's online services.	High
US 02	As a Customer , I want to log into my account, so that I can securely access my personal information and orders.	High
US 03	As a Customer , I want to manage my profile information, so that my details remain up to date.	Medium
US 04	As a Customer , I want to view the digital menu, so that I can see available food items and their prices.	High
US 05	As a Customer , I want to place a food order online, so that I can order food without waiting at the cafe.	High
US 06	As a Customer , I want to add, remove, and update food items in my cart, so that I can prepare my order before confirmation.	High
US 07	As a Customer , I want to confirm my order, so that the cafe can start processing it.	High
US 08	As a Customer , I want to view my previous orders, so that I can check my past purchases.	Medium
US 09	As a Customer , I want to provide feedback about my food and service, so that the cafe can improve its services.	Low
US 10	As a Customer , I want to reserve a table online, so that I can secure a table before visiting the cafe.	High
US 11	As a Customer , I want to check available tables, so that I can select a suitable time for my visit.	High

US 12	As a Customer , I want to track my order status, so that I know when my order is being prepared or is ready.	High
US 13	As a Customer , I want to view my bill or invoice, so that I can check the charges for my order.	High
US 14	As a Customer , I want to make payments online, so that I can conveniently pay for my order.	High
US 15	As a Customer , I want to cancel an eligible order and request a refund, so that I can recover my payment when cancellation is allowed.	Low
US 16	As a Customer , I want to receive notifications about my orders and reservations, so that I am informed about important updates.	Medium

Staff User Stories

ID	User Story	Priority
US 17	As a Cashier , I want to process customer payments, so that completed orders can be paid accurately.	High
US 18	As a Cashier , I want to generate bills and invoices, so that customers receive accurate payment records.	High
US 19	As a Cashier , I want to process approved refunds, so that cancelled eligible orders can be refunded correctly.	Low
US 20	As a Cafe Staff member , I want to update the status of an order, so that customers and other staff can know its current progress.	High
US 21	As a Cafe Staff member , I want to view active orders, so	High

	that I can prepare and serve them on time.	
US 22	As an Inventory Manager , I want to add, update, and remove inventory items, so that stock information remains accurate.	High
US 23	As an Inventory Manager , I want to monitor stock levels, so that I can identify items that need replenishment.	High
US 24	As an Inventory Manager , I want to receive low-stock alerts, so that I can reorder items before they run out.	Medium
US 25	As an Inventory Manager , I want to manage supplier information, so that I can easily contact and manage suppliers.	Medium
US 26	As an Inventory Manager , I want to record and manage purchases, so that inventory levels are updated correctly.	Medium
US 27	As a Supplier , I want to view purchase requests, so that I know which items the cafe needs.	Low
US 28	As a Cafe Staff member , I want to view the availability of required items, so that I can know which products can be prepared or served.	low
US 29	As a System Administrator , I want to assign roles and permissions, so that users can only access functions relevant to their responsibilities.	High

Manager / Admin User Stories

ID	User Story	Priority
US 30	As a Cafe Manager , I want to manage cafe staff information, so that employees can be	Low

	properly assigned and managed.	
US 31	As a Cafe Manager , I want to monitor daily cafe operations, so that I can ensure services run efficiently.	Medium
US 32	As a Cafe Owner , I want to view sales reports, so that I can understand the cafe's sales performance.	Medium
US 33	As a Cafe Owner , I want to analyze revenue, so that I can evaluate the financial performance of the cafe.	Medium
US 34	As a Cafe Owner , I want to identify best-selling food items, so that I can make better menu and business decisions.	Low
US 35	As a Cafe Manager , I want to view inventory reports, so that I can monitor stock usage and purchasing.	Medium
US 36	As a Cafe Manager , I want to view operational reports, so that I can identify problems and improve cafe operations.	Low
US 37	As a Cafe Owner or Manager , I want to view an interactive Power BI dashboard, so that I can understand business performance through visual analytics.	
US 38	As a Cafe Owner or Manager , I want to use business analytics and reports, so that I can make data-driven decisions.	

10. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a requirement or user story to be considered successfully completed.

ID	Related User Story	Acceptance Criteria
AC 01	US 01	Customer can register using valid required details; system validates the information; account is created successfully.
AC 02	US 02	Customer can log in using valid credentials; invalid credentials are rejected; authorized users are redirected to their account.
AC 03	US 03	Customer can view and update profile information; system validates updated details; changes are saved successfully.
AC 04	US 04	Customer can view available food items, prices, descriptions, and availability status.
AC 05	US 05	Customer can select food items and place an online order; system confirms the order successfully.
AC 06	US 06	Customer can add, remove, and change item quantities in the cart; cart total is updated correctly.
AC 07	US 07	Customer can review order details before confirmation; confirmed orders are recorded and sent for processing.
AC 08	US 08	Customer can view previous completed and cancelled orders with relevant order details.
AC 09	US 09	Customer can submit feedback; system validates the feedback and stores it successfully.
AC 10	US 10	Customer can select an available date and time and reserve a table; system prevents unavailable tables from being booked.
AC 11	US 11	System displays available tables based on the selected date, time, and required capacity.
AC 12	US 12	Customer can view the current order status; status is updated when staff changes the order progress.
AC 13	US 13	Customer can view an invoice containing ordered items, quantities, prices, taxes or charges, and total amount.
AC 14	US 14	Customer can select an available payment method; successful payments are recorded and linked to the order.
AC 15	US 15	Customer can cancel an eligible order within the allowed period; refund request is recorded and processed according to the refund rules.
AC 16	US 16	Customer receives notifications for important order and reservation updates.
AC 17	US 17	Cashier can process payments; system calculates the correct amount and records successful payment transactions.
AC 18	US 18	Cashier can generate a bill or invoice containing accurate order and payment details.
AC 19	US 19	Cashier can process only approved eligible refunds; refund status and amount are recorded correctly.
AC 20	US 20	Staff can update an order status; system saves the new status and makes it visible to authorized users and the customer.

AC 21	US 21	Staff can view active orders with required order details; orders can be identified and processed efficiently.
AC 22	US 22	Inventory Manager can add, update, and remove inventory items; system maintains accurate stock records.
AC 23	US 23	Inventory Manager can view current stock quantities and identify items below the required stock level.
AC 24	US 24	System generates a low-stock alert when an item's quantity reaches or falls below its defined threshold.
AC 25	US 25	Inventory Manager can add, update, view, and remove supplier information.
AC 26	US 26	Inventory Manager can record purchases; system updates the related inventory quantities correctly.
AC 27	US 27	Supplier can view assigned purchase requests and see required items and quantities.
AC 28	US 28	Staff can view the availability of required ingredients or items before preparing or serving products.
AC 29	US 29	Administrator can assign roles and permissions; users can access only functions permitted for their assigned role.
AC 30	US 30	Manager can add, update, view, and manage staff information.
AC 31	US 31	Manager can view key daily operational information such as orders, reservations, and staff activities.
AC 32	US 32	Owner can generate and view sales reports for a selected period; report values are based on recorded sales data.
AC 33	US 33	Owner can view revenue information for a selected period; displayed revenue values are calculated accurately.
AC 34	US 34	System identifies and displays best-selling food items based on recorded sales data.
AC 35	US 35	Manager can view inventory reports showing stock levels, usage, purchases, and related information.
AC 36	US 36	Manager can generate operational reports and identify relevant operational issues or performance trends.
AC 37	US 37	Owner or Manager can access the Power BI dashboard; dashboard displays relevant KPIs, charts, and business performance information.
AC 38	US 38	Owner or Manager can access business analytics and reports; information supports comparison, trend analysis, and data-driven decision-making.

11. Use Cases

Use cases describe how different users interact with the Bean & Brew Cafe Management System to achieve specific goals.

11.1 Main Actors

- Customer – Places orders, makes reservations, views order information, and provides feedback.
- Cashier / Staff – Manages orders, billing, and reservations.
- Inventory Manager – Manages stock and supplier-related information.
- Manager – Monitors operations and views reports.
- Administrator – Manages users, roles, and system configuration.

11.2 Use Case List

Use Case ID	Use Case	Primary Actor
UC 01	Register Account	Customer
UC 02	Login	Customer
UC 03	Manage Customer Profile	Customer
UC 04	View Digital Menu	Customer
UC 05	Manage Shopping Cart	Customer
UC 06	Place Food Order	Customer
UC 07	Confirm Order	Customer
UC 08	View Order History	Customer
UC 09	Track Order Status	Customer
UC 10	View Bill or Invoice	Customer
UC 11	Make Online Payment	Customer
UC 12	Cancel Order and Request Refund	Customer
UC 13	Submit Feedback	Customer
UC 14	Check Table Availability	Customer
UC 15	Reserve Table	Customer
UC 16	Receive Notifications	Customer
UC 17	Process Customer Payment	Cashier
UC 18	Generate Bill or Invoice	Cashier
UC 19	Process Refund	Cashier
UC 20	Manage Order Status	Cafe Staff
UC 21	View Active Orders	Cafe Staff
UC 22	View Item Availability	Cafe Staff
UC 23	Manage Inventory Items	Inventory Manager
UC 24	Monitor Stock Levels	Inventory Manager
UC 25	Manage Low-Stock Alerts	Inventory Manager

UC 26	Manage Supplier Information	Inventory Manager
UC 27	Manage Purchase Records	Inventory Manager
UC 28	View Purchase Requests	Supplier
UC 29	Manage Staff Information	Manager
UC 30	Monitor Daily Cafe Operations	Manager
UC 31	View Sales Reports	Manager
UC 32	View Revenue Analysis	Manager
UC 33	View Best-Selling Items	Manager
UC 34	View Inventory Reports	Manager
UC 35	View Operational Reports	Manager
UC 36	View Power BI Dashboard	Manager
UC 37	Perform Business Analytics	Manager
UC 38	Manage User Roles and Permissions	Administrator

11.3 Example Use Case – Place Order

Use Case ID: UC-03

Use Case Name: Place Order

Primary Actor: Customer

Preconditions:

- Customer is registered and logged in.
- Menu items are available.

Main Flow:

1. Customer opens the menu.
2. Customer selects available items.
3. Customer specifies quantities.
4. System adds selected items to the cart.
5. System calculates the order total.
6. Customer reviews the order.
7. Customer confirms the order.
8. System creates the order.
9. System displays an order confirmation.
10. Staff receives the new order.

Alternative Flow:

- If an item becomes unavailable, the system informs the customer and prevents that item from being ordered.
- If the cart is empty, the system does not allow the customer to submit the order.
- Postcondition:
- A valid order is stored in the system.
- The order is available for staff processing.

12. Assumptions

1. The following assumptions are made when developing the Bean & Brew Cafe Management System:
2. Customers have access to a smartphone, tablet, or computer with an internet connection when using online functions.
3. Cafe staff have basic computer knowledge and can use the system after appropriate training.
4. The cafe has a stable internet connection for accessing the web-based system.
5. Authorized staff members will maintain accurate menu, inventory, reservation, and supplier information.
6. Inventory quantities entered into the system are assumed to be accurate.
7. The cafe will define appropriate user roles and access permissions.
8. The cafe will define table capacities, reservation time slots, and cancellation policies.
9. The cafe will define minimum stock levels for inventory items.
10. Management will use system-generated reports to support operational decision-making.
11. The system will initially support the operational requirements of the selected cafe and may be expanded in the future.

13. Constraints

The development and implementation of the Bean & Brew Cafe Management System are subject to the following constraints:

1. Time Constraint – The system must be developed and delivered within the allocated project period.
2. Budget Constraint – Development must be completed within the available project budget and resources.
3. Resource Constraint – The project has limited development personnel and technical resources.
4. Technology Constraint – The system is dependent on the selected software technologies, database, hosting environment, and available development tools.
5. Internet Dependency – Online ordering, reservations, and other web-based functions require a stable internet connection.
6. Data Availability Constraint – The quality of reports and inventory information depends on the accuracy and completeness of data entered into the system.
7. Integration Constraint – Integration with external payment gateways, banking systems, delivery platforms, and third-party services may be limited within the current project scope.
8. Security Constraint – The system must protect customer, staff, transaction, and business information while operating within the security capabilities of the selected technologies.
9. Scope Constraint – Features outside the defined project scope cannot be included unless the project scope is formally reviewed and approved.
10. User Adoption Constraint – The effectiveness of the system depends on staff and management adopting the new digital processes and using the system correctly.